

DSOR: A Scalable Statistical Filter for Removing Falling Snow from LiDAR Point Clouds in Severe Winter Weather

Akhil Kurup¹, Logan Schexnaydre¹, Chaz Cornwall¹, and Jeremy Bos¹

Abstract—Falling rain and snow introduce noise in LiDAR returns resulting in both false positive and false negative object detections. In this article, we present the Dynamic Statistical Outlier Removal (DSOR) filter, a statistical PCL-based filter capable of removing snow with a higher recall than the state-of-the-art statistical snow de-noising filter while being 28% faster and with lower time complexity. We also compare DSOR to a CNN-based filter and find again it has improved recall. Owing to its improved recall DSOR is especially relevant in situations where obstacles are embedded in snow and where GPUs may be undesirable for cost, complexity, or energy use concerns.

DSOR is available at
https://bitbucket.org/autonomymtu/dsor_filter.

I. INTRODUCTION

Autonomous Vehicle (AV) perception systems often rely on Light Detection And Ranging (LiDAR) sensors for perception, mapping, and localization. Precipitation such as rain and snow can reduce the performance of these systems [1]. LiDAR sensors are particularly affected due to their inherent beam divergence and short pulse duration. For many AV-focused LiDAR sensors, snowflake detection noise is concentrated near the sensor and manifests as a clutter noise as seen in Fig. 1. Snow clutter noise can introduce false detections as well as obscure important obstacles leading to critical false-negative detections [2]. As an example, Fig. 1 (top) shows a LiDAR point cloud captured in Hancock, MI on the 12th of February 2020 with a snow rate of 0.6 in/hr (1.5 cm/hr). Two oncoming vehicles are obscured by the snow and evade Autoware’s Euclidean clustering for object detection [3].

A common technique for dealing with noise in LiDAR point clouds is a filtering step prior to a detection stage. LiDAR returns are volumetrically non-uniform and follow a log-normal distribution [5]. This results in a point density that decreases with increasing range. Generalized filters, however, do not account for the spatially varying distribution of snow and, therefore, risk losing important key points in the environment. To address this limitation, we present the Dynamic Statistical Outlier Removal (DSOR) filter to remove snow from LiDAR point clouds. Our solution dynamically changes the filter threshold with range to remove most of the snow and outlier noise while preserving environmental features compared to other statistical filters.

Fig. 1 (bottom) shows most of the snow has been cleared by our DSOR filter to reveal a pair of oncoming vehicles that are obscured in the top image. We show that our

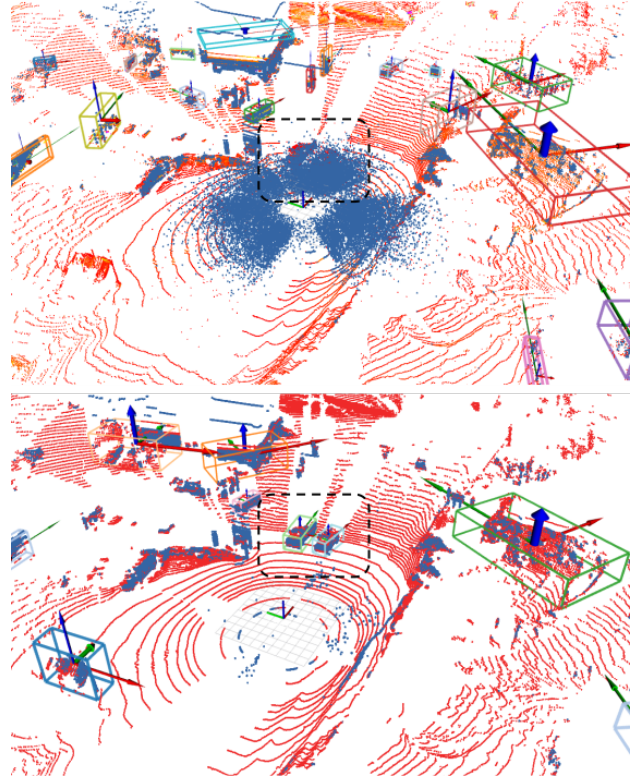


Fig. 1: Part of our Winter Adverse Driving dataSet (WADS) [4] showing moderate snowfall 0.6 in/hr (1.5 cm/hr). (top) Clutter from snow particles obscures LiDAR point clouds and reduces visibility. Here, two oncoming vehicles are concealed by the snow. (bottom) Our DSOR filter is faster at de-noising snow clutter than the state-of-the-art and enables object detection in moderate and severe snow.

filter improves visibility for tasks such as object detection and semantic scene understanding in inclement weather. Localization is explicitly not addressed in this work.

We compare our filter to the Dynamic Radius Outlier Removal (DROR) filter [6], which we consider to be the state-of-the-art filter for statistical snow de-noising. DSOR filter is also shown to have a 4% increase in recall and operate 28% faster relative to the DROR filter. More importantly, DSOR is demonstrated to have $O(N \log(N))$ time scaling with the number of points compared to $O(N^{2/3})$ for DROR. As a point of comparison,

DSOR is also compared to WeatherNet [7] a CNN based de-noiser intended for use in fog and rain but trained here on our winter weather dataset, WADS [4]. This comparison

¹Electrical and Computer Engineering, Michigan Technological University, Houghton, MI, USA

is intended to be purely exemplary as statistical filters differ from CNN approaches in several important ways. The CNN is constrained to operate on data with the same dimensions as the training data, which would prevent a CNN from working with LiDAR that has varying scanning patterns or other changes to the input data. The necessity of a GPU to ensure real time operation is another. That said, both our version of WeatherNet and other approaches [8] achieve higher precision but lower recall than DSOR. In our testing, DSOR has a 14% higher recall than WeatherNet while running 32% faster on a CPU.

The contributions of this paper can be summarized as follows:

- We introduce the DSOR filter to de-noise point clouds corrupted by falling snow.
- We show that the DSOR filter outperforms the state of the art statistical filter in recall and has a lower time complexity resulting in improved scalability.
- We show the recall of the statistical filters evaluated here is at least on par with CNNs.

II. RELATED WORK

Automotive LiDARs can be adversely affected by weather-related scattering and absorption effects. Rain adds noise to point clouds severely degrading the performance of object detection algorithms [9]. Falling snow in winter can be detected as diffuse, solid objects, often smaller than the laser beam cross-section. Rasmussen *et al.* [10] show that small snow particles act as Lambertian targets and create significant backscattering. To better understand the interaction of snow with automotive 3D LiDAR, Roy *et al.* [11] modeled the interaction between snowflakes and laser pulses. They show that snow detections are concentrated near the sensor and detection statistics are governed by the characteristics of the laser beam and rate of precipitation. Michaud *et al.* [5] show that the probability of detecting snow in automotive 3D LiDAR follows a log-normal distribution with distance from the sensor.

Noise introduced by snow particles in LiDAR scans can be filtered out in a number of ways; [1] and [11] introduce intensity-based filters. These filters assume LiDAR returns from snowflakes will have a lower intensity and remove snow noise by intensity-value thresholding. However, intensity is dependent on the laser wavelength, distance, and target reflectance, and as shown in [10], backscatter from snow particles near the sensor can be significant. To overcome this limitation, Park *et al.* [1] propose a second clustering stage based on point density to preserve objects of interest. However, in heavy snow with a high volume density, such techniques will classify snow as an object of interest.

Another filtering technique is to apply image processing on LiDAR point clouds. Here, point clouds are converted to a 2D representation, and techniques like Gaussian filters or their derivatives are applied to remove noise. However, as shown by Charron *et al.* [6], 3D point clouds are sparse and such approaches fail to remove snow effectively and have the negative effect of smoothing edges in keypoint features.

Consequently, it is reasonable to assume that recall may be preferred over precision in some perception pipelines. Duan *et al.* [12] use Principal Component Analysis (PCA) to convert 3D point clouds to 2D and apply DBSCAN clustering to find sparse regions and remove them. However, as mentioned previously, clustering techniques fail at high snowfall rates.

Several CNN-based de-noising approaches have also emerged in recent works. Heinzler *et al.* [7] and Yu *et al.* [8] show that Neural Networks trained on datasets with inclement weather can achieve high precision in removing snow. However, these methods first convert a 3D point cloud to a 2D image which introduces some limitations. 3D points that are behind other objects in space, either get removed or stacked with the same point, in the image. This, again, leads to a loss of key points and errors during re-projection. Milioto *et al.* [13] propose a kNN post-processing step to mitigate some of these errors, but this does not completely address the issue.

Finally, there is an argument to be made that LiDARs here are only a single element in a perception system and, therefore, a simple filter with high recall when fusing with other sensors will provide the best information to downstream object detection systems while saving on energy usage. Given that the aim of this work and others is snow-noise removal it would seem recall would be the more performance-relevant metric.

We apply filtering to the 3D points directly using a statistical approach. The Point Cloud Library (PCL) [14] includes general-purpose filters such as the voxel grid filter, Statistical Outlier Removal (SOR) filter, and Radius Outlier Removal (ROR) filter; these are, however, not designed for snow. They fail to account for the non-uniformity of the point clouds and hence Charron *et al.* [6] propose the Dynamic Radius Outlier Removal (DROR) filter for removing snow corruption. Their filter dynamically changes the search radius with range to remove snow effectively. Wang *et al.* incorporate the intensity of the LiDAR return in their Dynamic Distance-Intensity Outlier Removal (DDIOR) filter [15].

In this work, we compare our DSOR filter with the DROR filter and the WeatherNet CNN. All approaches use our Winter Adverse Driving dataSet (WADS) [4] for training and testing. A portion of WADS features semantically labeled LiDAR point clouds with both falling and fallen snow during moderate to heavy snowfall rates.

III. FILTER IMPLEMENTATION

PCL's SOR filter is a general noise removal filter widely used for pre-processing point clouds. However, it does not account for the non-uniform distribution of the point cloud, and when applied to a scan with falling snow, fails to remove it as seen in Fig. 2 (second from left).

The state-of-the-art DROR filter applies a threshold to the mean distance of points to their neighbors in a given radius to remove sparse snowflakes. To address changing LiDAR point spacing with distance, the DROR filter changes the search radius as the distance increases from the sensor. The

DROR filter achieves a high accuracy but fails to produce a clean point cloud at high snowfall rates as shown in Fig. 2 (second from right).

Algorithm 1 Dynamic Statistical Outlier Removal (DSOR) filter

Input: Point Cloud ($\mathbf{P}$) = $p_i, i = 1, \dots, N; p_i = (x_i, y_i, z_i)$
 $\mathbf{k}$ = minimum number of nearest neighbors
 $\mathbf{s}$ = multiplication factor for standard deviation
 $\mathbf{r}$ = multiplication factor for range

Output: Filtered Point Cloud ($\mathbf{F}$) = $f_i, i = 1, \dots, N; f_i = (x_i, y_i, z_i)$

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1:  $P \leftarrow KdTree$ 
2: for  $p_i \in P$  do
3:    $mean\_distances \leftarrow nearestKSearch(\mathbf{k})$ 
4: end for

5: calculate: mean ( $\mu$ )  $\leftarrow mean\_distances$ 
6: calculate: standard deviation ( $\sigma$ )  $\leftarrow mean\_distances$ 
7: calculate: global threshold ( $T_g$ )  $\leftarrow \mu + (\sigma \times \mathbf{s})$ 

8: for  $p_i \in P$  do
9:    $distance \leftarrow \sqrt{x_i^2 + y_i^2 + z_i^2}$ 
10:  calculate: dynamic threshold ( $T_d$ )  $\leftarrow T_g \times \mathbf{r} \times distance$ 

11:  if  $mean\_distances < T_d$  then
12:     $f_i \leftarrow p_i$  {Inliers}
13:  end if
14: end for

15: return  $\mathbf{F}$ 

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A. The DSOR Filter

The DSOR filter is an extension of PCL's SOR filter, designed to address the inherent non-uniformity in point clouds. Algorithm 1 helps visualize our approach. The point cloud is first loaded into a k-d tree and the k-nearest neighbor search is applied for every point to accumulate user-provided k neighbors. The mean (μ) and standard deviation (σ) of the mean distances are calculated to compute a global threshold given by equation 1.

$$T_g = \mu + (\sigma \times constant) \quad (1)$$

To account for the non-uniform spacing of points, the distance of every point from the sensor is calculated and a new *dynamic threshold* (T_d) is computed using equation 2

$$T_d = T_g \times r \times distance \quad (2)$$

Points with mean distances less than T_d are inliers and are retained. The detected outliers, corresponding to the snow noise are removed. Here, r is a constant, a multiplicative factor for point spacing. Increasing r will increase T_d and will result in the filter rejecting fewer points, whereas,

reducing r will decrease T_d resulting in a more aggressive filter. The DSOR filter, when applied to a snow-corrupted point cloud, produces the cleanest point cloud as compared to the other approaches as shown in Fig. 2 (rightmost).

IV. EVALUATION AND RESULTS

The DROR, WeatherNet CNN, and DSOR filters are used on data collected during heavy snow events with severely reduced visibility. The DROR and DSOR filters have been implemented on an Intel® Core™ i7-9750H CPU with 32 GB of RAM. In this section, we compare the performance of the statistical filters using both qualitative as well as quantitative results; WeatherNet results are included for reference. Qualitative results include both the visual performance of the statistical filters and the distribution of filtered points as a function of range. To quantify the results alongside a CNN benchmark, the precision and recall as well as the filtering time have been considered. Results from the statistical filters have been averaged over 100 point clouds with moderate falling snow. Throughout this study, parameters for the DSOR filter had the following values: $k = 5$, $\sigma = 0.01$, & $r = 0.05$.

A. WeatherNet CNN

To compare the performance of the statistical filters against a Convolutional Neural Network (CNN), the WeatherNet CNN was selected as the example model [7]. WeatherNet takes a range image for distance and intensity as an input. This CNN was implemented, trained, and tested in Python using the WADS dataset [4]. Approximately 1300 and 100 labeled frames were allocated to train and test the model, respectively. WeatherNet uses a Cross Entropy Loss and Adam optimizer with a learning rate of 0.001 to train on the dataset. LiSnowNet, was published by Yu *et al.* [8] after the completion of this work. We note that our snow-focused version of WeatherNet is comparable in filtering performance to LiSnowNet.

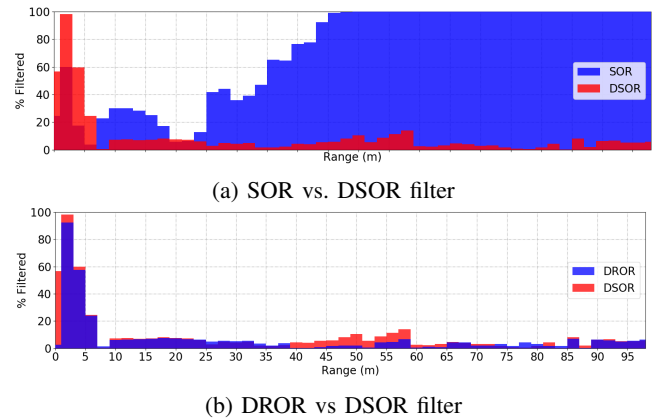


Fig. 3: Percentage of filtered points as a function of range (averaged over 100 point clouds). The DSOR filter outperforms both the SOR and DROR filters in ranges $< 20\text{m}$ where most of the snow is concentrated.

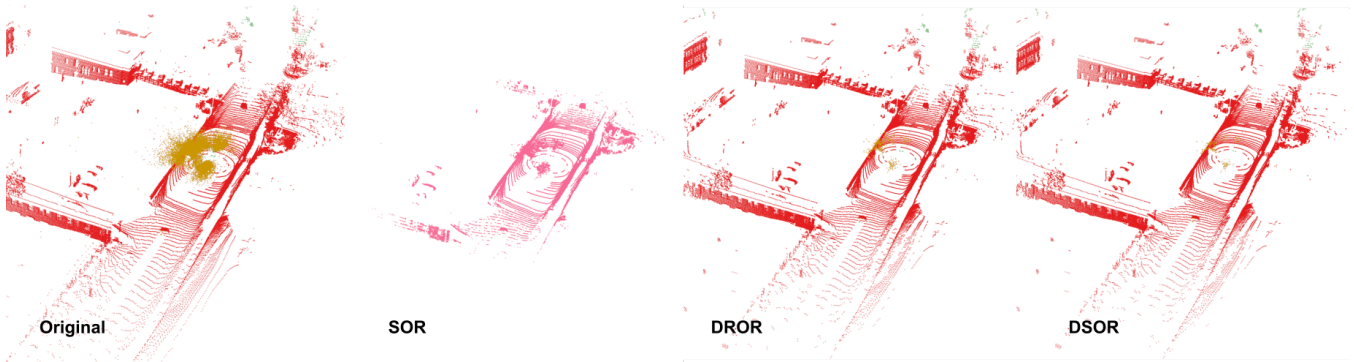


Fig. 2: Qualitative comparison of the SOR filter, state-of-the-art DROR filter, and our DSOR filter (left to right). The original point cloud shows snow clutter (orange points) that degrades LiDAR perception. The DSOR filter removes more snow compared to both the SOR and DROR filters and preserves most of the environmental features.

B. Qualitative evaluation

Fig. 2 shows a qualitative comparison of the original point cloud, corrupted by snow, passed through different filtering techniques. The SOR filter fails to remove most of the snow and removes key environmental features. The DROR filter removes most of the snow while preserving environmental features. The DSOR filter removes most of the snow from the point cloud while preserving key points removed by the SOR filter.

C. Quantitative evaluation

To understand the visual performance of the filters, the percentage of filtered points as a function of the range has been plotted in Fig. 3.

DSOR vs SOR filter: Fig. 3a shows that the DSOR filter removes significantly more points than the SOR filter in the first 20 meters where most of the LiDAR interaction with snow particles occurs. Beyond 20 meters, the DSOR filter removes significantly fewer points than the SOR filter thus preserving key environmental features.

DSOR vs DROR filter: Fig. 3b shows that in the first 20 meters, the DSOR filter also removes more points than the DROR filter. However, beyond 20 meters, the DSOR filter removes more points potentially leading to a lower precision at range.

TABLE I: Precision and Recall of the DSOR filter compared with the SOR, DROR, and WeatherNet filters (averaged over 100 point clouds).

Filter	Precision [%]	Recall [%]
DSOR	65.07	95.6
DROR	71.51	91.89
WeatherNet	94.41	81.92

D. Precision and Recall

In the context of this work, the aim of a filter should be to remove as much snow as possible (recall) and leave other points untouched (precision). It is possible to filter out all of the snow (recall of 100%) at the expense of

environmental features (low precision) using any filter. Such a filter, however, will not be very practical. Both the DSOR filter and the DROR filter have been tuned to filter most of the snow and preserve as much of the environment as possible as shown in Fig. 2.

As shown in Table I, the DSOR filter achieves a greater recall (95.6%) than the DROR filter (91.9%). The DSOR also has a higher recall than the WeatherNet CNN. This shows that the DSOR filter removes more of the snow corruption correctly than the DROR filter. The DROR filter, however, achieves a greater precision (71.5%) at filtering snow as compared to the DSOR filter (65.1%). This shows that the DSOR filter removes more non-snow points or environmental features than the DROR filter.

From Fig. 3b it is evident that the loss in precision for the DSOR filter stems from removing points beyond 20 meters. At this range, we observe that features such as tree tops, poles, and overhanging power lines have been filtered out. Such features are sparse, and in the winter can return just a few points.

TABLE II: Execution time of the DSOR filter compared with the SOR, DROR, and WeatherNet CNN filters (averaged over 100 point clouds).

Filter	Execution Time		DSOR Speedup (ms)
	mean (ms)	std. dev. (ms)	
DSOR	369.68	18.1	-
SOR	371.98	19.2	2.3
DROR	510.55	33.9	140.87
WeatherNet	499.66	16.58	129.98

E. Filter Rate

Table II compares the time taken to filter a point cloud by the DSOR, SOR, DROR, and WeatherNet CNN filters. Results have been averaged over 100 point clouds collected by a 64-channel LiDAR with roughly 200k points each. The DSOR filter is, on average, 141 ms or 28% faster than the state-of-the-art DROR filter. The DSOR filter is also 129.98 ms or 32% faster than the WeatherNet CNN.

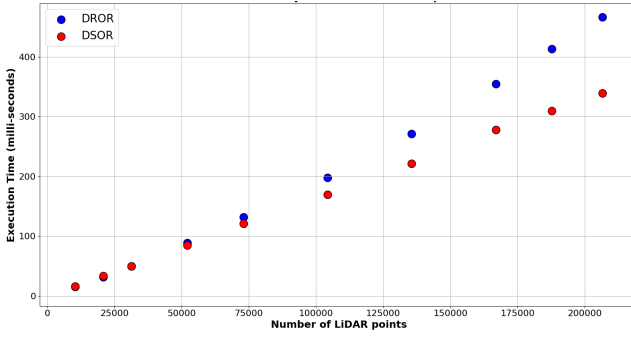


Fig. 4: Relative time complexity of the DSOR filter compared to the DROR filter with the size of a point cloud. The DSOR filter scales better than the state-of-the-art DROR filter.

F. Scalability

As LiDARs continue to evolve, sensors with higher point density and number of lasers are expected. State-of-the-art AVs also concatenate returns from multiple LiDARs for increased visibility and dense spatial sampling. This increases the number of points being captured and processed in the pipeline. For this reason, filters need to scale with the size of point clouds to ensure adaptation on AV systems. Using the CNN in this study would require retraining for each down-sampled point cloud; therefore, the CNN was not included.

To compare the time taken for filtering by the DSOR filter and DROR filter, multiple levels of sub-sampling were applied to a sequence of point clouds from WADS [4]. Fig. 4 shows that the DSOR filter has a lower execution time as compared to the DROR filter as the size of a point cloud increases. The DROR filter uses a radius search and we speculate that the DROR filter has an approximate time complexity of $O(N^{2/3})$ [16]. A curve fit on Fig. 4 is consistent with this assessment. The DSOR filter uses a k-nearest neighbor search and has a lower time complexity of $O(N \log(N))$ implying that our filter will scale better with the size of LiDAR point clouds.

V. CONCLUSION

Winter weather negatively affects perception systems used in autonomous vehicles. In particular, LiDAR point clouds suffer from false detections (both positive and negative) introduced by falling snow. To address this shortcoming we have presented the Dynamic Statistical Outlier Removal (DSOR) filter for de-noising snow from LiDAR point clouds. We show that our filter outperforms the state of the art DROR filter by achieving a 4% higher recall and 28% lower execution time, without optimization. Our filter also has a 14% higher recall than WeatherNet. The DSOR filter has a lower time complexity enabling it to scale better with the size of point clouds. Further improvements in performance may be realized by using fast clustering and voxel-subsampling as shown in [17].

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